

Debodeep Banerjee

PhD Researcher in Artificial Intelligence University of Pisa & University of Trento

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🔗 [Debodeep94](#) 🌐 [Website](#)

RESEARCH PROFILE

PhD researcher working on hybrid human-machine learning and decision-making under uncertainty. My work focuses on integrating vision-language models, statistical guarantees such as conformal prediction, and human feedback into intelligent systems. I am particularly interested in designing AI frameworks that support real-world decision processes.

RESEARCH INTERESTS

Large-language models Vision-language models Computational Social Science/ AI in Politics Hybrid decision-making systems Statistical machine learning

EDUCATION

PhD in Artificial Intelligence	2022–Present University of Pisa & University of Trento
Master in Data Science	2022 Sapienza University of Rome
M.Sc. in Statistics	2018 University of Madras
B.Sc. in Statistics (Hons)	2016 Asutosh College, University of Calcutta

PUBLICATIONS

Conference and Workshop Papers

[\[Paper\]](#) Banerjee, D. et al. *Learning to Guide Human Decision Makers with Vision-Language Models. Accepted at International Conference on Discovery Science, 2026.* Proposes Learning-to-Guide, a novel human-AI decision-making paradigm where vision-language models generate interpretable guidance to support human experts rather than replace them.

[\[Paper\]](#) Banerjee, D. et al. *MedGellan: LLM-Generated Medical Guidance to Support Physicians. HLDM Workshop at ECML-PKDD 2025.* Develops an annotation-free framework that leverages LLMs to generate clinical guidance from medical records for assisting physician decision making.

[\[Paper\]](#) Banerjee, D. et al. *Hybrid Decision Making via Conformal VLM-generated Guidance. HW&HAI at HHAI 2026.* Introduces ConfGuide, a framework that combines conformal risk control with vision-language models to generate targeted medical guidance with statistical guarantees.

[\[Paper\]](#) Banerjee, D. et al. *TradeVerse: A Longitudinal Benchmark of Political Negotiation in International Trade Under review at The Forty-First AAAI Conference on Artificial Intelligence.* Tradeverse turns decades of WTO trade disputes into a benchmark for tracking multi-round, multi-party political trading negotiation

RESEARCH EXPERIENCE

PhD Researcher	2022–Present University of Pisa & University of Trento
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Research on hybrid AI systems combining vision-language models, conformal prediction, and human guidance mechanisms.

Data Science Intern

Jun–Oct 2022 Project Consulting SRL, Rome

Developed AI-based methods for video data analysis.

AI Intern

Jul 2021–Jan 2022 Baker Hughes (Nuovo Pignone Tecnologie), Italy

Worked on applied machine learning solutions in industrial settings.

Research Intern

May–Jun 2017 Indian Statistical Institute, India

RESEARCH PROJECTS

Nonparametric Modal Regression for Directional Responses Master Thesis, Sapienza University of Rome Supervisor: Prof. Pierpaolo Brutti

Developed EM-based modal regression algorithm for directional predictors and linear responses.

Classical Inference on Single Binomial Proportion M.Sc. Thesis, University of Madras Supervisor: Dr. Subbiah M.

Studied coverage probability and expected run length properties of confidence interval procedures.

PRESENTATIONS

International Conference on Theory and Application of Statistics and Information Science (2018) Bayesian analysis of $1 \times j$ contingency table

National Conference at Marian College (2018) CSR and Sustainable Development in India

TECHNICAL EXPERTISE

Programming: Python, R, SQL

Machine Learning: NLP, Vision–Language Models, Statistical Learning

Gen-AI: RAG, Ollama, LangGraph, vLLM, prompt engineering, agentic-AI

Statistics: Inference, regression modelling, conformal prediction

Tools: Git, LaTeX

Visualisation: matplotlib, plotly, QGIS — *creator of a data-visualisation initiative communicating Indian social statistics to a public audience through visual storytelling.* (DESIDATA)

AWARDS

First Place — Inter-Departmental Quiz, University of Madras

First Place — Inter-Collegiate Quiz, Madras Christian College

LANGUAGES

English (Fluent) Bengali (Fluent) Hindi (Fluent) Italian (Basic)